

MTA Bus Fare Evasion

Overview

General Description

The Metropolitan Transportation Authority (MTA) is a public-benefit corporation responsible for public transportation in the state of New York serving 12 counties in southeastern New York, along with two counties in southwestern Connecticut under contract to the Connecticut Department of Transportation (CDOT). The MTA is the largest transportation network in North America.

Bus service within New York City is operated by MTA agencies New York City Transit (NYCT) and MTA Bus Company (MTABC).

The bus fare evasion (BFE) rate is the estimated percentage of riders who illegally board buses. Understanding BFE is important for estimating bus ridership and gauging lost revenue. The data is used regularly by NYCT's Office of Management and Budget (OMB), and Division of Operations Planning (OP).

Since Q4 2020, the BFE rate has been estimated using Automated Passenger Counter (APC) and Automated Fare Collection (AFC) data. Until Q1 2020, fare evasion was estimated using surveys conducted by traffic checkers, with fare evasion on Select Bus Service (SBS) routes, except the S79, being estimated based on data from enforcement surges by the EAGLE Team. Fare evasion was not reported in Q2 or Q3 2020, as checkers were not deployed due to the COVID-19 pandemic.

The BFE rate began being publicly reported in MTA Board meeting materials in April 2019. This dataset includes the rate for local and limited-stop (LCL/LTD) buses, Select Bus Service (SBS) buses, express (EXP) buses and the systemwide total.

This dataset was published during the first phase of the MTA's commitment to increasing transparency. We continually examine all our published and publishable data with a view to both providing datasets that can be effectively utilized by our customers and the public at large, and to providing regular, automated updates to these datasets efficiently and sustainably. Consequently, this dataset may be restructured and/or combined with other similar datasets in the future.

Data Collection Methodology

Current Methodology

The current methodology estimates the BFE rate using APC and AFC data, including from SBS sidewalk vending machines. Unlike previous methods, it is calculated based on actual data, not a survey.

Processing APC data

APCs are infrared sensors mounted above each door on a bus that record the number of boardings and alightings each time that doors open and close. Validation efforts have showed APC counts were correct within +/- 1 passenger at 99 percent of stops when comparing manual observations of video recordings. Not all buses are equipped with APCs, and this coverage also varies by borough. All new buses are equipped with APCs; coverage should increase as old buses are retired or retrofitted with readers.

APC records also include the bus id, a timestamp, and a GPS ping for the geolocation. This data is written from the buses to a production database, which usually happens in real-time. If the bus is out-of-network, it may wait until it arrives back at the depot to do a bulk upload using depot WiFi. 99.8 percent of APC data is available by end of day.

Each morning, OP gets stop-level data from APCs for the previous day, and an automated procedure aggregates the data into bus trips. Bus trips are defined by the first APC boarding and the last APC alighting. In order to better define comparisons to AFC data, nearby trips are grouped together into blocks. Trips are grouped together into blocks if there are fewer than 15 minutes between the end of one trip and the start of the next trip on a given bus. This is done since riders may board during one trip then alight during the next. Without grouping, both trips are imbalanced, with the first trip having one additional boarding, and the second having an additional alighting.

The process also indicates which trips are unbalanced (have a large difference between the numbers of ons and offs on each trip). The process compares the number of balanced APC trips from the previous month with the total number of trips that ran based on processed BusTime data. Data from routes with less than one percent APC coverage were filtered out until 2024 due to some changes in the process, which are [detailed below](#).

Processing AFC data

MetroCard transactions and OMNY tabs data is obtained from ridership data. Data comes in each morning with yesterday's numbers. Before Q2 2024, this data was provided by the Office of

Management and Budget (OMB). Data had come with a six-week lag, and OP generated preliminary numbers for a given month about 10 to 20 days after the end of the month. At that time, the numbers from the previous month were finalized.

For each block, the process counts and matches OMNY and MetroCard fare data with APC boardings in the same time interval for the trip, with a buffer of seven minutes added at the start of the interval and a buffer of one minute added to the end of it. While OMNY data timestamps are rounded down to the nearest second, MetroCard data can only be rounded down to the nearest six-minute interval. The time before and after the start of the trips helps account for this rounding, and any time synchronization issues between the AFC/OMNY and APC systems.

An estimated percentage of riders paying in cash is also included to determine a route's total paid ridership. Cash data is first taken from data collected by fareboxes. It is collected by the farebox in a trip record file separate from MetroCard data that does not retain each transaction, but summarizes it into loosely defined trips, though this has proved to be unreliable. These loosely defined trips are therefore not used to distribute the payments, and instead cash data is aggregated at the depot level. Cash data contains both the count of riders boarding with cash and revenue from these transactions, and includes different fare types, including reduced and full fares. Since cash is received on 15th or 16th of each month, the previous month's numbers are used prior to that. Cash payments vary by less than 0.5 percent.

Split fare boardings are captured via MetroCard data, while cash short-drop boardings are determined by applying a blended fare derived from the share of full and reduced fares in total cash boardings. This blended fare was calculated in 2015 separately for NYCT and MTA Bus using data from 2014, with the share of revenue from reduced fares being 8.1 percent for NYCT, and 11.5 percent for MTA Bus. This base cash data-both boardings and revenue-is gathered at the depot level, and for each depot, it is multiplied by the ratio of the actual cash collected to the amount that the data file reported.

To distribute cash ridership from the depot level to the route level, the estimated ratio of cash ridership for each route at the depot, based on older OP fare evasion surveys, is used. In these fare evasion surveys, short-drop riders were counted as fare evaders.

Later, during the BFE calculation, to obtain total paid ridership, the percentage of riders on each bus route paying in cash, the cash rate, is calculated. It is calculated as actual cash divided by cash fare. Since this cash ridership figure includes revenue from partially paid boardings, or short-drop fares, this

formula counts short-drop riders as equal to a much-reduced number of effective full fare riders. Since the bus fare evasion surveys used to distribute cash ridership from depots to bus routes counted short-drop riders as fare evaders, and the BFE calculation counted short-drop riders as partial riders, there is a small double-counting when including effective paid riders.

The APC data distributed short-drop riders between the paid and fare evader category based on effective payment, with short-drop MetroCard fares being included as paid riders, while short-drop cash fares were included as evaders.

APCs are not able to count children under 44 inches tall, are not able to differentiate between short-drops and valid fares, and cannot distinguish authorized nonpayment, which include questionable or legal boarding without a fare paid, from evasion. To account for authorized non-payment, the systemwide fare evasion rate calculated is subtracted by 1.7 percent. OMB obtained this figure by using fare evasion survey data between March 2019 and February 2020 as a baseline and multiplying this figure by 30 percent. This reflected the fact that APCs do not count children under 44 inches tall, which made up 70 percent of the category in data before October 2019, when this category stopped being measured. This calculation accounted for changes in short-drop ridership by using cash revenue market share as a variable, and by calibrating the short-drop converting rate.

Ridership is removed on fare-free bus routes (both for the Q70, which is fare-free permanently, routes that were part of the fare-free pilot between September 2023 and August 2024 (Bx18A/B, S46/S96, B60, M116, Q4)) for the calculation of fare evasion, resulting in an approximately 0.3 percent difference in the overall system fare evasion rate.

Calculation

The BFE rate for a given route is the difference between the number of boardings and paid fares divided by boardings; this calculation is only performed on trips with APC. To get the systemwide BFE rate, each route's BFE rate is weighted by its total paid ridership. Each route's BFE rate is the summation of the BFE rate for each trip block. Paid ridership for a trip block, i , is equal to the sum of the number of MetroCard and OMNY transactions matched to each trip block divided by the cash adjustment factor (*cash rate*) for the route and day type for that block.

Here are the formulas for paid fares for a trip block, and for the fare evasion rate for a bus route:

$$Paid\ Fares_i = \frac{(MetroCard_i + OMNY_i)}{(1 - cash\ rate)}$$

$$BFE_{route} = \sum_{i=1}^n \frac{((APC \text{ Boardings})_i - (Paid \text{ Fares})_i)}{(APC \text{ Boardings})_i}$$

Where:

i = trip block,

n = number of trips/trip blocks with valid APC and AFC data in the month,

$cash \ rate$ = cash adjustment factor for the route and day type
(weekday/Saturday/Sunday) of trip/trip block i ,

$(APC \text{ Boardings})_i$ = number of boardings from APCs on trip/trip block i ,

$MetroCard_i$ = number of MetroCard transactions matched to trip/trip block i , and

$OMNY_i$ = number of OMNY transactions matched to APC trip/trip block i ,

Next, the initial borough fare evasion rate for routes with APC data for a month is calculated. Until 2024, when fewer buses had APCs, only routes with at least 1 percent of trips covered by APCs were included in the calculation for m . The fare evasion rate of each route is multiplied by paid ridership for each route divided by 1 subtracted by the fare evasion of that route. This is then multiplied by 1 divided by the summation of ridership on each bus route divided by 1 minus the fare evasion rate for each route. This bus fare evasion rate for routes with APC data is finally obtained by summing this value for each bus route.

The formula for this calculation is:

$$BFE_{boro-APC} = \sum_{route=1}^m BFE_{route} \times \frac{riders_{route}}{1 - (BFE_{route})} \times \frac{1}{riders_{boro-APC}}$$

Where:

m = # of bus routes,

BFE_{route} = the fare evasion rate of each route,

$riders_{route}$ = the official paid ridership for each route that month, and

$$riders_{boro-APC} = \sum_{route=1}^m \frac{riders_{route}}{1 - (BFE_{route})}$$

The next step is to estimate the final boroughwide nonpayment rates. Until 2024, when not all routes had at least 1 percent of trips covered by APCs, the first step was assuming that routes without adequate APC coverage had the same fare evasion rate as routes with APC coverage ($BFE_{boro-APC}$). Next, the fare evasion rate for a route is multiplied by official paid ridership for a route divided by 1 minus the fare evasion rate for a route. Then, it is multiplied by 1 divided by the summation of paid ridership for each route multiplied by 1 plus the fare evasion rate for a route. This value is then summed across all routes in the borough.

This formula is:

$$BFE_{boro} = \sum_{\substack{route \\ e=1}}^k BFE_{route} \times \frac{riders_{route}}{1 - BFE_{route}} \times \frac{1}{riders_{boro}}$$

Where:

k = # of routes in the borough, and

$$riders_{boro} = \sum_{route=1}^k riders_{route} \times (1 + BFE_{route})$$

To calculate the systemwide bus fare evasion rate, the same formula from the previous equation is used, except with the summation being of the five borough rates. The formula for systemwide ridership, $riders_{system}$, is different than that for ridership in a borough, $riders_{boro}$, being the summation of $riders_{boro}$ divided by 1 subtracted by the fare evasion rate for the five borough rates.

The formula for the systemwide APC nonpayment rate is then:

$$BFE_{system} = \sum_{boro=1}^5 BFE_{boro} \times \frac{riders_{boro}}{1 - BFE_{boro}} \times \frac{1}{riders_{system}}$$

Where:

BFE_{boro} = the fare evasion rate of borough b ,

$riders_{boro}$ = the total ridership in borough b , and

$$riders_{system} = \sum_{boro=1}^5 \frac{riders_{boro}}{1 - BFE_{boro}}$$

Changes

Starting in March 2023, the MTA changed its methodology for measuring the SBS fare evasion rate to a method consistent with that used to measure evasion on local and limited buses. This methodology is believed to be more accurate as it is based on a wider sample. The change in methodology resulted in an increase in the evasion rate for SBS. Data for this methodology is available back to Q4 2020.

Details on the previous methodology used to measure the SBS evasion rate are available in the [next section](#).

In 2024, the methodology was updated to make use of a pipeline that runs daily and automatically instead of being done manually once a month, and to better reflect actual fare evasion, though there were only minimal changes from existing numbers. The main changes are listed below:

The previous method from OP dropped routes with low APC coverage when aggregating from route-level to borough-level evasion, or from borough-level to system-level. The other routes were aggregated and weighted based on their number of trips. The new method just aggregates all trips that had APC coverage. This is believed to better represent actual fare evasion, as routes with low coverage tend to be higher in fare evasion. Since the process of matching payments to trips, and therefore routes, relies on the quality of geospatial/time data from those payments, which can be noisy, paid ridership is now scaled up somewhat to account for the small (<2 percent) of payments that are not matched to a route.

The process now removes fewer APC trips during data cleaning as a procedure was developed that does not underestimate BFE when riders stay on for multiple trips, which is most common on loop routes. In addition, the updated Bus Matching procedure identifies a greater number of trips, especially those with more abnormal run patterns, providing a larger sample for estimating BFE.

Prior to 2025 Q2, buses with zero MetroCard payments throughout an entire day of service were removed from fare evasion calculation. This accounted for buses not having AFC data downloaded due to maintenance, despite customers actually paying in line with the APC boardings. As OMNY ridership share grew, this was deemed likely to exclude actual cases of buses with no MetroCard payments in a day. This filter is no longer used.

Additionally, prior to 2025 Q2, routes with an origin at Canarsie-Rockaway Parkway Station (B6, B42, B82) were excluded from calculation due to the free transfer within fare control at which customers did

not typically use the fare box upon entry. Due to changes at the station that require payment, these are now included in the calculation.

In September 2025, paid ridership totals were updated to exclude \$0 split fare transactions and bus fare evasion rates were recalculated back to Q1 2022 using the current methodology to reflect these new totals. \$0 split fare transactions averaged approximately 30K daily rides from 2022-2025 (around 4% of bus MetroCard ridership, 2% of overall bus paid ridership). As MetroCard ridership declined throughout 2025, the number of these transactions also declined. Removing these transactions from paid ridership totals resulted in a slight increase in estimated fare evasion rates of approximately 1 percentage point.

Previous methodology

Select Bus Service

Fare evasion on all SBS routes except the S79 was estimated differently as off-board fare payment was used for these routes, meaning traffic checkers would not be able to know which riders paid without asking to see their tickets. The BFE rate was obtained based on data from enforcement surges by the EAGLE Team in which plainclothes inspectors boarded a bus, had all doors closed, and asked to see proof of payment for all passengers. Fare evasion was calculated as the number of riders without valid proof of payment divided by the number of passengers inspected. This approach was developed with a 95 percent confidence level, and an error range of +/-2.6 percent.

Non-SBS routes

Background

Before COVID19, for all bus routes except express bus and SBS routes other than the S79 SBS, systemwide fare evasion was estimated using surveys conducted by traffic checkers riding randomly sampled bus trips. Quarterly monitoring using this method began in Q2 2012 by OP, following the completion of fare evasion studies in 2008 and 2011. This method stopped being used due to the COVID-19 pandemic to minimize staff present on buses.

MTA Audit Services and the MTA Inspector General had recommended the use of technology to replace or supplement in person checks since it was challenging for checkers to listen to the farebox, categorize each boarding, and monitor multiple doors while facing multiple potential distractions. An MTA Audit Services review had found that the survey, in some cases, significantly undercounted fare evasion.

The sample for the surveys also excluded school trips (extra bus service added to the schedule to service school dismissal) and only included full path trips, excluding mid-point reliefs. Checks were not done on

trips starting overnight since, with more infrequent service, no more than one trip could be scheduled per shift. Trips had to have scheduled arrival times no later than 1 a.m., and departures no earlier than 5:15 a.m. for Manhattan and Queens, 5:45 a.m. for the Bronx and Brooklyn, and 6:15 a.m. for Staten Island.

The measurement of fare evasion on express buses was removed from the survey in 2014 since the rate was relatively low in surveys completed in 2012 and 2013 and since the cost of surveying was very high since checkers required test cards, and due to many one-way trips. All steps detailed that apply to each borough also formerly applied to express buses.

Step 1. Determine the desired margin of error.

Fare evasion surveys, like any statistical survey, are only point estimates of the true population values. Since doing a 100 percent sample, with every single bus boarding in the system being surveyed, is impossible, a random sample of bus trips was used to observe the systemwide BFE rate. With random samples inherently having some error since the characteristics of the sampled population are not the same as those of the entire population, and since the results would be subject to data collection errors, some level of error, known as the margin of error (MOE), has to be determined as acceptable. For example, at a fare evasion rate of five percent, the error rate has to be lower than five percent in order for the results to be meaningful. A lower error margin, at a given statistical level of confidence, implies a larger sample size.

This MOE is calculated using a formula based on the sample size, sample type, and the variability within the survey results. This MOE is then used to create a confidence interval around the point estimate, tied to a 95 percent probability that the true population value is within that range.

Formula used between 2012 and 2019

Until early 2019, the MOE was calculated using a formula for estimating the MOE for a population proportion with a simple random sample. With this formula, the MOE was equal to the z-score for the selected 95 percent confidence interval of 1.96 multiplied by the square root of the estimated proportion of paying riders to all riders multiplied by the evasion rate (1 minus that proportion) divided by the number of bus boarding observed by traffic checkers. The formula is:

$$ME = z \sqrt{\frac{p^{\wedge} (1 - p^{\wedge})}{n}}$$

Where:

z – z-score (confidence interval). We assume a 95% confidence interval (z-score = 1.96),

$p^{\wedge}$ – estimated proportion of observations of paying riders to all riders, and

n – number of observed bus boardings

Knowing any two variables allowed for the third one to be calculated. Historical data provided guidance to the values of all three variables.

Formula used between 2019 and 2020

In early 2019, the way the MOE was calculated was revised. The previous formula had been based on a simple random sample, which would require observations of random individual passengers, as opposed to random bus trips or pairs of trips, which is what was surveyed. To be more accurate, the MOE calculation was changed to treat round trips as clusters of observations. The MOE formula for estimating a proportion in a one-stage cluster sample would be used, similar to the formula that began to be used for the subway between Q4 2018 and Q4 2020.

The evasion rate for the new formula is calculated as the summation of the number of selected persons with the attribute of interest (illegal entries) in a specific cluster divided by the number of total selected persons or entries (legal + questionable + illegal) for that specific cluster. These calculations are done using the following formulas:

$$p = \sum_{i=1}^N \frac{a_i}{m_i}$$

$$m_i = L_i + Q_i + a_i$$

The MOE is calculated with a 95 percent confidence level using the formula for the confidence interval of a proportion, which is the z-score for the 95 percent confidence level (1.96) multiplied by the square root of the variance of the proportion of illegal entries to total entries across sampled, or the fare evasion rate, $(v(p))$. The numerator for the variance of the proportion is the summation of the number of selected persons with the attribute of interest (illegal entries) in a specific cluster subtracted by the evasion rate for that survey multiplied by the number of total selected persons or entries (legal + questionable + illegal) for the specific cluster to the second power. The denominator for the variance of the proportion is the number of sampled clusters, n , multiplied by the number of sampled clusters minus 1 multiplied by the average number of selected persons (entries) per cluster to the second power. These calculations are done using the following formula:

$$ME = 1.96\sqrt{v(p)} = 1.96\sqrt{\frac{\sum_{i=1}^N (a_i - pm_i)^2}{n(n-1)\bar{m}^2}}$$

Where:

a_i is the number of illegal entries in sampled cluster i ,

m_i is the number of entries in sampled cluster i ,

L_i is the number of legal entries in sampled cluster i ,

Q_i is the number of questionable entries in sampled cluster i ,

$v(p)$ is the variance of p ,

p is the proportion of total number of illegal entries to total entries across sampled clusters,

n is the number of sampled clusters, and

$\bar{m}^2$ is the average number of entries per sampled cluster

Step 2. Determine the minimum number of observations to reach the MOE.

The next step was to determine the minimum number of observations to reach the error margin. It was assumed that the expected evasion rate was not very different from the previous quarter or year's rate, which allowed historical data to be used to determine the required number of observations for each next cycle of the study. The number of observations for each borough could be taken directly from the number of observations already made for the previous cycle as the MOE in the historic data was low enough to be used.

The fare evasion rate is taken from fare evasion observations from the year ending at the end of the bus operator pick (bus operators select, or pick, their work assignments, which correspond with schedule changes, about four times a year, and each schedule is known as a pick) from one year before, and the error margin taken from Step 1, are plugged into the MOE formula to calculate the minimum number of observations, n . The next step was to calculate the number of observations for each borough. Bus routes were assigned to strata based on the borough assigned in the bus route name. For example, sampled trips from the Q59, an interborough bus route between Queens and Brooklyn, were assigned to the strata for Queens.

In early 2019, when the margin of error calculation for bus fare evasion was changed, the sample generation process was revised to be based on trip counts as opposed to ridership, and this step was changed. One reason for the change was that the previous method meant that any borough with a

disproportionately low or high number surveys was under- or over-represented in the total systemwide result. This was because survey results were added together without any weighting, regardless of whether the number of surveys completed was proportional to the ridership in a given borough.

Between 2012 and early 2019, the number of observations for each borough was calculated by weighing the systemwide n by borough ridership in the previous year, obtained from OMB. Between early 2019 and 2020, this was done by weighing the systemwide n by the scheduled number of local and limited bus trips in each borough, also including MTA Bus trips.

Step 3. Determine the minimum number of bus trips.

The next step would determine the minimum number of bus trips to be surveyed. Different methods were used before and after 2019.

Method used between 2012 and 2019

Between 2012 and early 2019, the next step was to convert sample observations to sample trips since the random sample consisted of bus trips and not observations (individual boardings). The minimum number of trips to be sampled for each borough was obtained by dividing observations, n , from the previous step, by the load factor. The load factor is the average number of observations per trip and was calculated for each borough by dividing total ridership in the borough by the total number of bus trips for the previous year.

Method used between 2019 and 2020

With the change in early 2019, the number of round trips chosen for each pick would initially be based on survey completion in the last bus operator pick. The targeted number of round trips chosen would be the number of round trips completed in the previous pick divided by 80 percent, to account for traffic checker availability.

Step 4. Divide trips proportionally into weekday, Saturday, and Sunday categories.

In the following step, the number of trips was apportioned among the days of the week (weekday, Saturday, Sunday). While different load factors were calculated for borough, this was not done for day type. Trips were divided proportionally to ridership between 2012 and early 2019, and by total number of trips between 2019 and 2020.

Method used between 2012 and 2019

The number of trips to be sampled in each combination of borough and day type was obtained by multiplying the number of trips during that period by the percentage of all ridership in a week for a borough for that specific day type.

The formulas for these calculations are:

$$\text{Weekday Trips} = \text{Trips} * \text{Weekday Borough Ridership} / \text{Total Borough Ridership}$$
$$\text{Saturday Trips} = \text{Trips} * \text{Saturday Borough Ridership} / \text{Total Borough Ridership}$$
$$\text{Sunday Trips} = \text{Trips} * \text{Sunday Borough Ridership} / \text{Total Borough Ridership}$$

Method used between 2019 and 2020

In early 2019, the sample size for each borough was changed to be in proportion to the share of bus trips in the borough, and the sample size for each day of the week was changed to be proportioned to the share of bus trips on those days instead of based on ridership. As such, the number of weekday trips to be sampled was obtained by multiplying the number of trips to be sampled by 5/7. For both Saturday and Sunday trips, the number of trips to be sampled was multiplied by 1/7.

Step 5. Generate random sample.

Next, the sample size was scaled up or down in proportion to the number of traffic checkers and the length of the pick. For example, with six traffic checkers working five days a week, doing two round trips per day, and 13 weeks in a quarter, there would be about 800 round trips that could be sampled. The sample size for each category would be obtained by breaking down the scaled sample size by borough and day type. The final step was to generate a random sample of trips representative of the above distribution about every quarter, typically four times per year-once for each new bus operator pick and occasionally between picks.

Samples were done around picks since trips in one pick might not exist in the next one, even if in the same month. Since calendar quarters do not usually align with each pick, OP aggregated surveys from multiple picks to create quarterly BFE estimates. It was not believed that the small number of observations around the borders of reporting periods would make a major difference in the results.

Sample generation was done by giving a unique number to each bus route ID/trip ID combination and using a random number generator to pick trips. To ensure enough trips in each category, the procedure used buckets for each of the combinations of day type and borough. The procedure ran until every bucket had the requested number of trips.

Sample assignment

Each week, OP analysts would schedule survey work for traffic checkers. Surveys would be assigned in a relatively balanced manner using a sorted list of trips in a database with not all of a time of day or given borough done first. They would also regularly review which surveys were not completed to reschedule

them for another chance to be completed. This database sorted trips-both those scheduled and those not surveyed-inversely weighted by strata so trips in undersampled strata would appear first. This would make it more likely for the analyst to assign those trips and correct any imbalance.

Traffic checkers were generally scheduled to observe two round trips per day, though sometimes only one round trip was possible-especially when it was a long route starting from the checker reporting location. These trips would be assigned to be in the same borough, when possible. Each sampled trip was paired with a return trip on the same route for practical reasons, meaning about half the surveyed trips came from the random sample. This round trip would also be paired with another based on how close the origin of the two round trips were to each other to reduce travel time between assignments.

Conducting the surveys

Traffic checkers would ride on assigned bus trips near the front door and record all boardings using a custom-built app on tablets. However, if there were technical issues, they were required to perform surveys on paper. Checkers would count boardings and assign them to categories by listening to the distinct tone the farebox produced to indicate an invalid MetroCard or incomplete fare payment and watching each passenger's interaction with the farebox and driver. Since bus operators would also not always press the reset button, short drops were hard to detect.

BFE was divided into two categories during observations: illegal evasion and authorized non-payment, which included questionable or legal boarding. Illegal fare evasion included all situations where a person clearly did not pay their fare as required. It includes passengers who board a bus without making an attempt to pay the fare or paying less than required (Short Drop). Authorized nonpayment was not included in the number of illegal boardings. Questionable boarding included instances where it was unclear whether a person paid or needed to pay, such as riders who gave pieces of paper to the bus operator, like SBS tickets or paper transfers. Legal boarding includes riders who are not required to pay, such as children under 44 inches tall, uniformed officers, organized school groups with a letter presented to the operator, and employees who show but do not dip their employee pass.

Checkers would also note unusual occurrences. If there was a service disruption on a bus, including its deviating from its planned route, terminating before its scheduled end of the route, becoming too crowded, or having farebox problems, the checker could optionally exit the bus at an intermediate stop and board the next bus in the same direction on the same route and resume the survey.

After surveys were completed, the data was uploaded to a central database. An analyst would run data validation reports to identify any data the checker may not have collected correctly or that might have been corrupted during the upload process. Multiple automated checks would flag potential issues for analysts to examine. The analyst could exclude any checks with inaccurate or corrupted data.

Other Changes

Starting in July 2019, MTA Bus, NYCT bus, and subway fare evasion observations were performed by the same group of traffic checkers, allowing more checkers to fully devote their time to observing fare evasion. Previously, fare evasion for MTA Bus had been studied and calculated separately, though with the same procedures as NYCT. The studies were combined with random cluster samples generated together.

Multiple efforts were introduced in fall 2019 to improve the accuracy of the data. The number of supervisors was increased from 1 to 4, and a new automated report was introduced to highlight potential issues with checks during data validation. A minimum completion goal of 80 percent was also implemented. In October 2019, fare evasion surveys were simplified to allow checkers to better focus on identifying evasion, with a reduced number of categories for type of evasion.

To reduce undercounts of evasion from the third door of articulated buses due to sightline problems when they were crowded, staff were added to monitor that door, and a pilot was begun in late 2019 to monitor it using video. Every month, OP analysts would provide a list of articulated buses that were surveyed to the Traffic Checker Supervisor to request videos of the third door from various depots. Once the video file was acquired, traffic checkers would watch the videos and tally the number of evaders entering through the third door. Rear door video observations were used to create an average scaling rate on a monthly basis to estimate rear door observations based on second door observations. This rate was calculated by finding the sum of the rear door evaders and third door evaders and dividing third door evaders by rear door evaders, calculating the ratio of the third door to the second door evaders.

Statistical and Analytic Issues

This dataset includes the BFE rate for local and limited-stop buses back to Q1 2019, express buses back to Q4 2020, and for SBS back to Q1 2021. The Q1 2020 BFE survey was incomplete in March due to the COVID-19 pandemic. There were no measurements in Q2 or Q3 2020 for any type of bus service due to the pandemic, and since no local bus fares were collected between April and September.

Since data between Q1 2019 and Q1 2020 was obtained using surveys conducted by traffic checkers, human error could take place. Data after Q4 2020 is from the newer APC methodology. Some minor changes were made to this methodology in 2024 and 2025.

Once reported, fare evasion records are not updated. Note that the “true” underlying fare evasion number for a given quarter may have changed due to late arriving data (buses may rarely return from maintenance to service and report AFC or APC data several months after service). We do not expect these to affect the final numbers by more than 0.1%, and to maintain consistent reporting these adjustments are not made. The exception to this is the update made in Q4 2025, when historical values were recalculated back to Q1 2022 to reflect the new approach to short-drop payments.

Limitations of Data Use

There are no limitations on the data at this time.

Release Notes

Version 1.0.0 release notes instituted and methodology tweaked (08/20/2024)

Version 2.0.0 ridership field removed and systemwide fare evasion records added (01/29/2025)

Version 2.0.1 documentation updated (03/13/2025)

Version 2.0.2 documentation updated and historical data recalculated (10/24/2025)